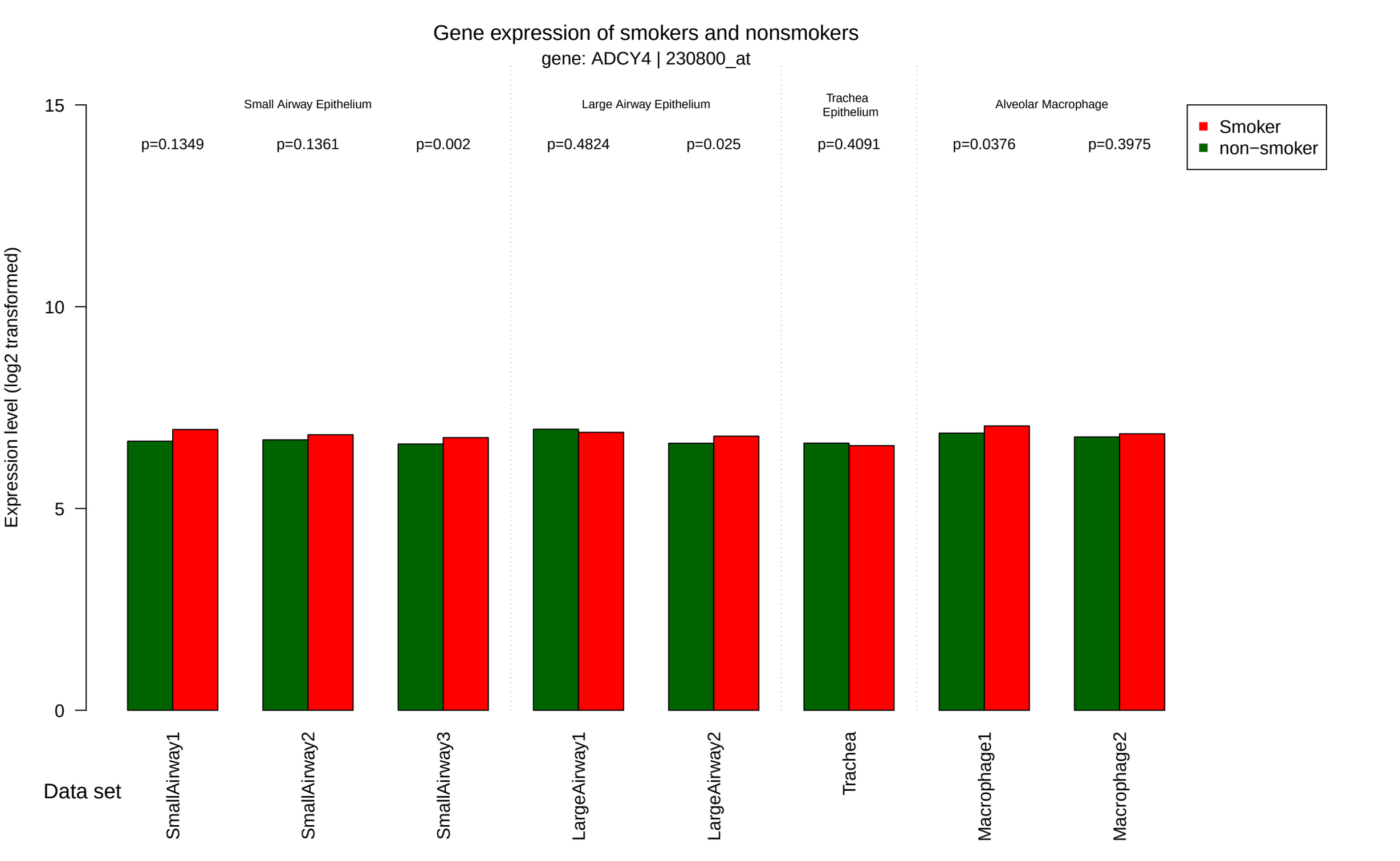
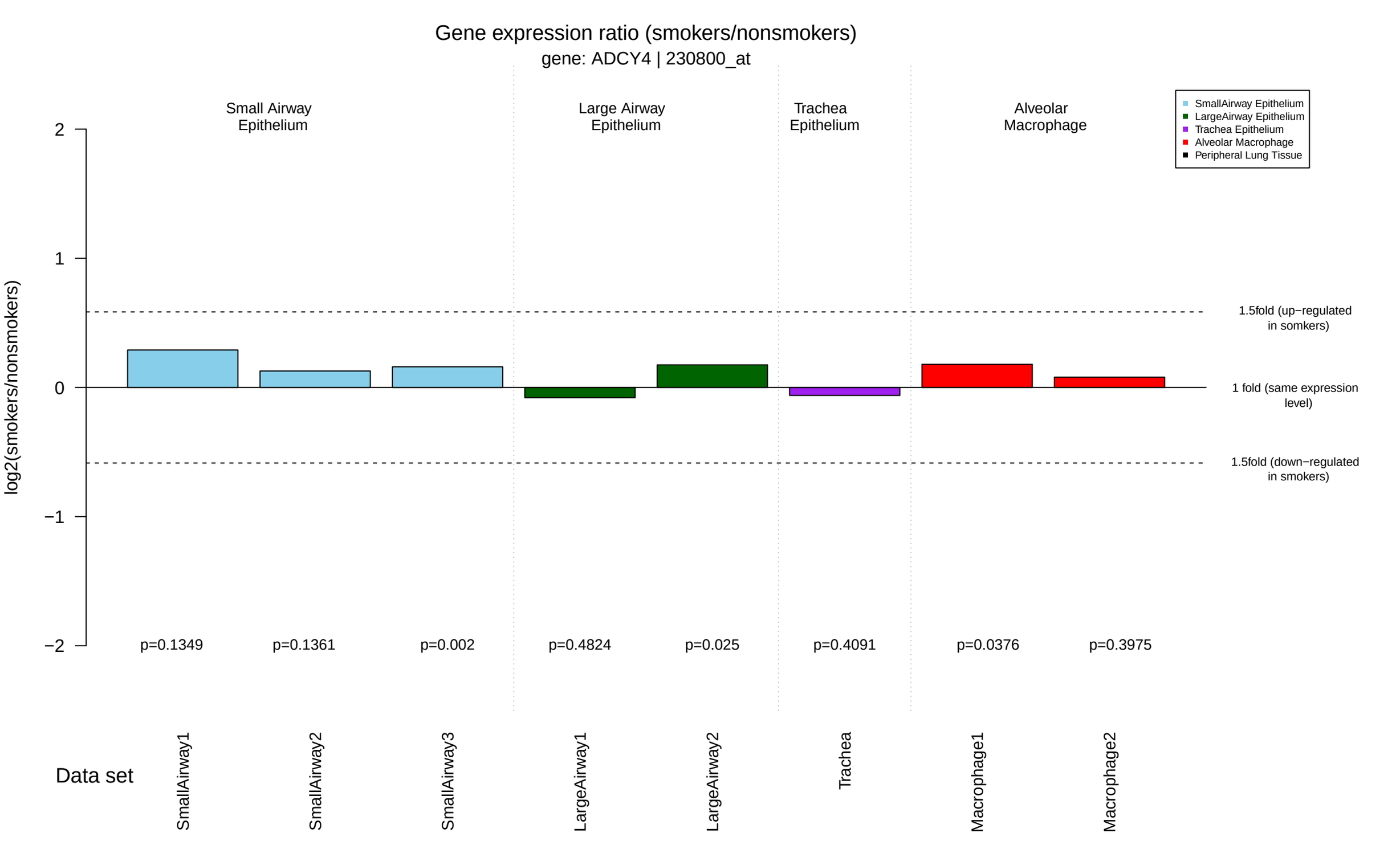
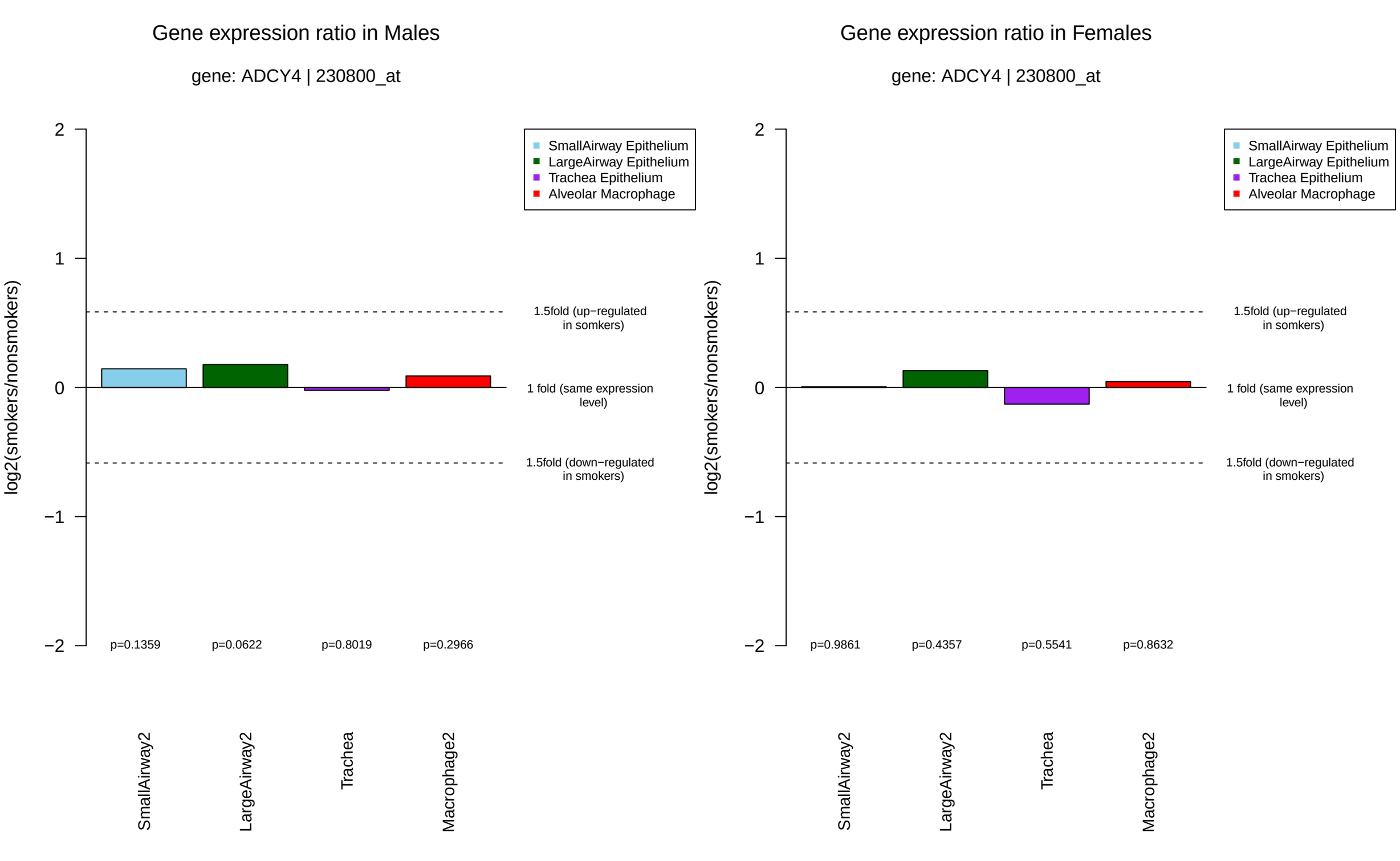
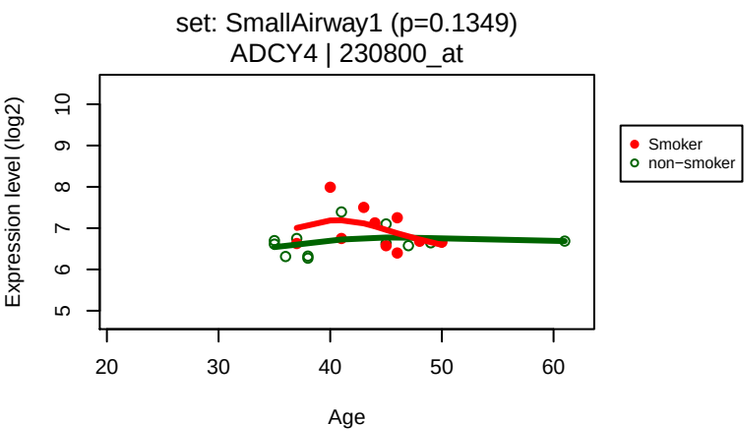
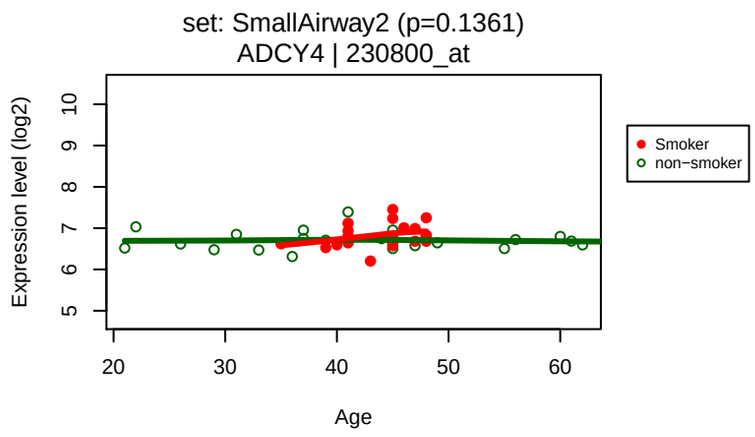
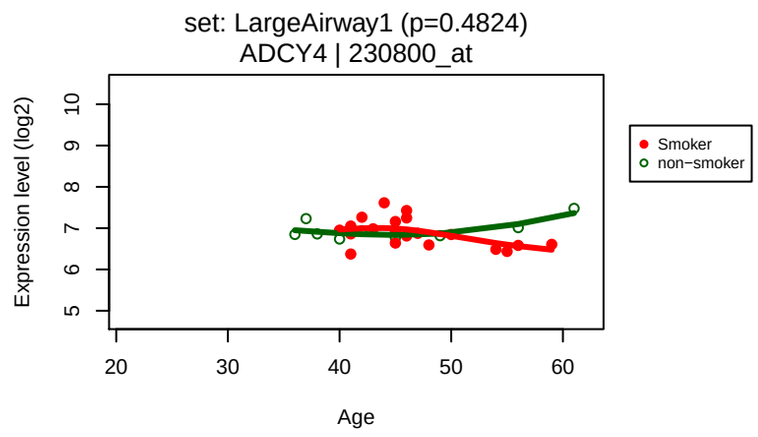

Gene expression of smokers and nonsmokers
gene: ADCY4 | 230800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1349
p=0.1361
p=0.002
p=0.4824
p=0.025
p=0.4091
p=0.0376
p=0.3975
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

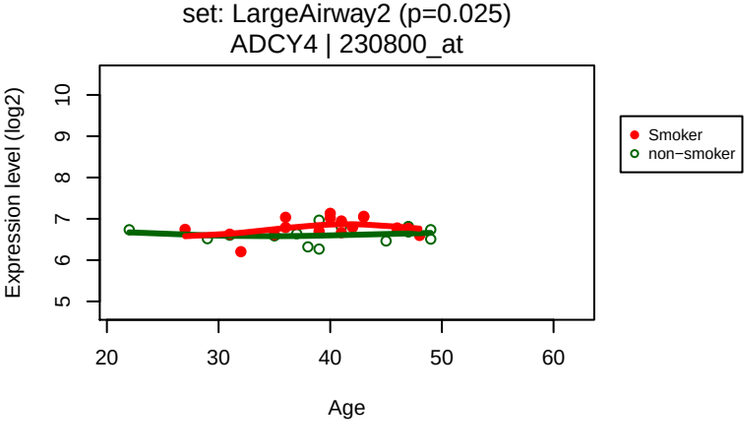

Gene expression ratio (smokers/nonsmokers)
gene: ADCY4 | 230800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1349
p=0.1361
p=0.002
p=0.4824
p=0.025
p=0.4091
p=0.0376
p=0.3975
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

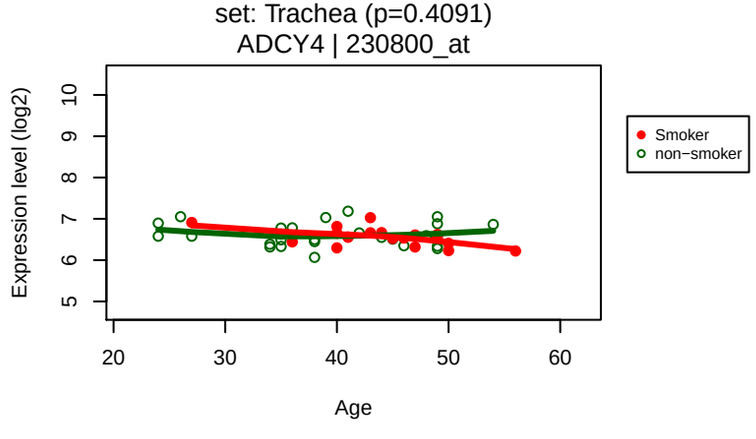

Gene expression ratio in Males
gene: ADCY4 | 230800_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1359
p=0.0622
p=0.8019
p=0.2966
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ADCY4 | 230800_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9861
p=0.4357
p=0.5541
p=0.8632
SmallAirway2
LargeAirway2
Trachea
Macrophage2

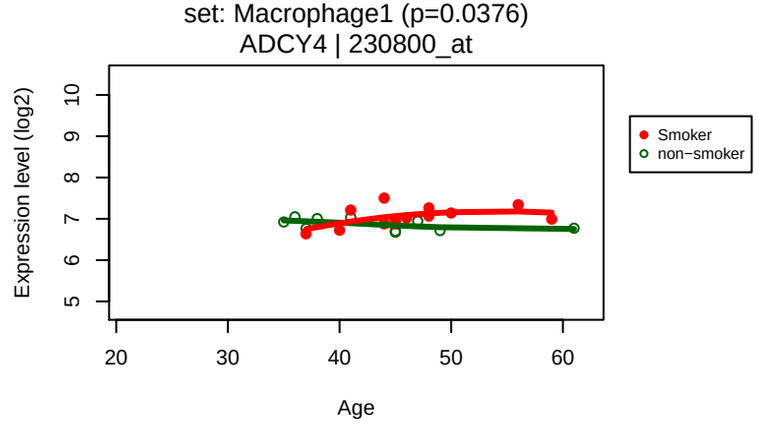

set: SmallAirway1 (p=0.1349)
ADCY4 | 230800_at
Expression level (log2)
Age
Smoker
non−smoker

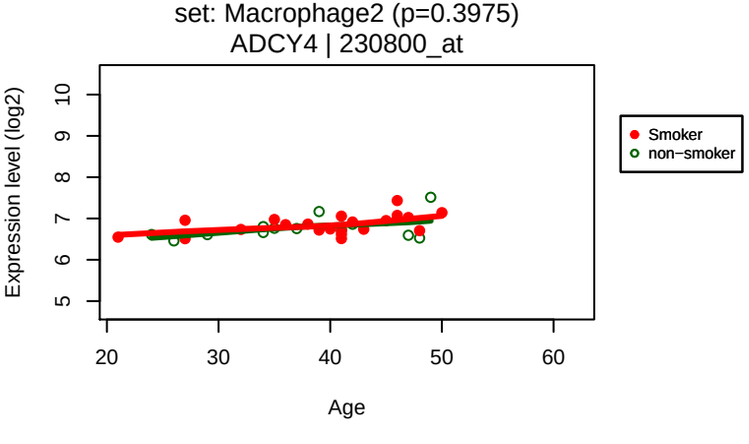

set: SmallAirway2 (p=0.1361)
ADCY4 | 230800_at
Expression level (log2)
Age
Smoker
non−smoker

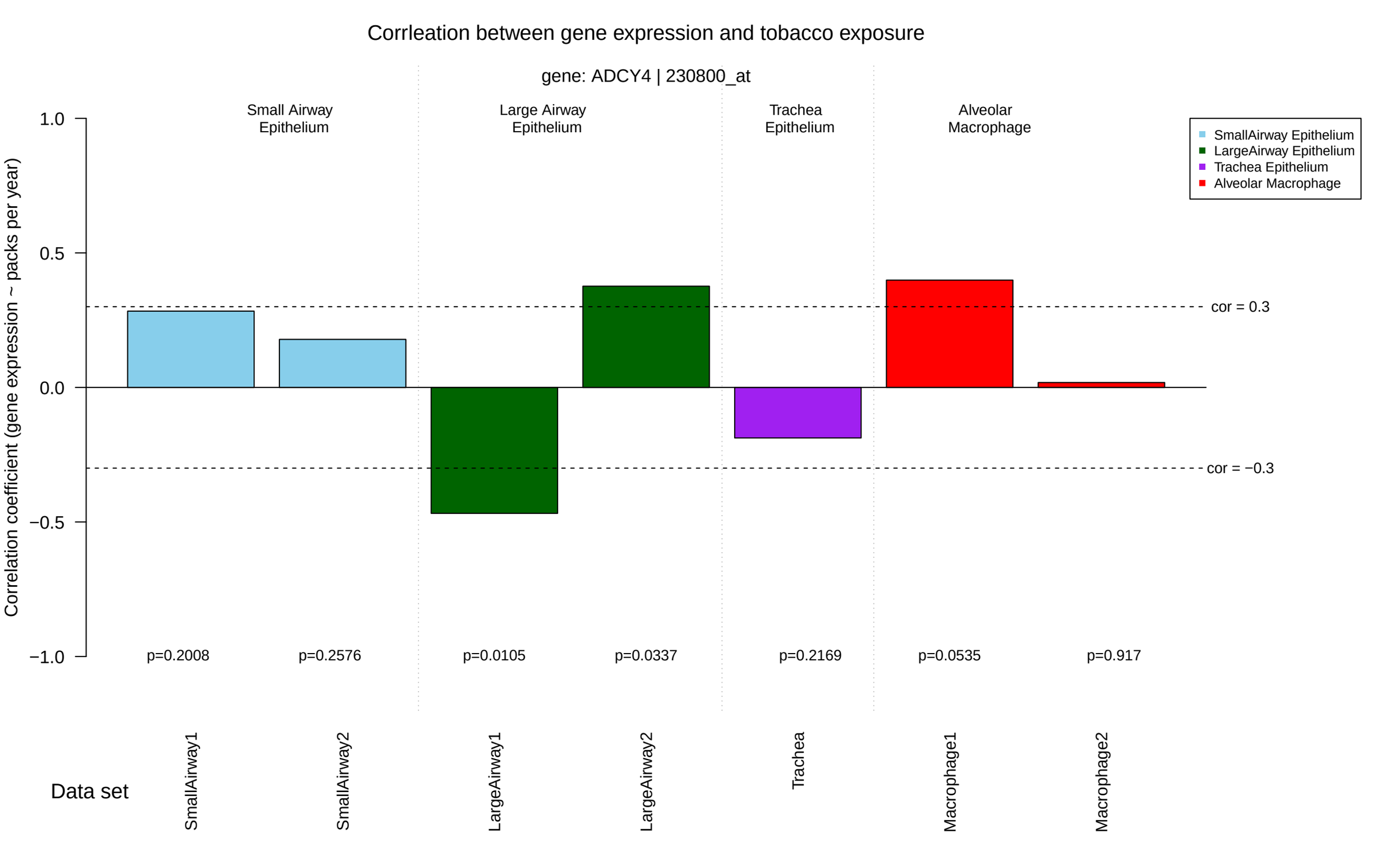

set: LargeAirway1 (p=0.4824)
ADCY4 | 230800_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.025)
ADCY4 | 230800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4091)
ADCY4 | 230800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0376)
ADCY4 | 230800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3975)
ADCY4 | 230800_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ADCY4 | 230800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2008
p=0.2576
p=0.0105
p=0.0337
p=0.2169
p=0.0535
p=0.917
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2